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PAPER_956: Spectral Ladder Phonon Mapping (26-Level)

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** et_{phonon_resonance}.py §7 (SpectralLadderPhononMapping) **Calculator:** SpectralLadderPhononMappingCalc (CP4 #540) **CVW:** v2.0.0 compliant

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Abstract

We construct the phonon mapping for the 26-state HRes spectral ladder, converting each energy level $E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}$ to its corresponding phonon frequency ω_n and quality factor Q_n. The mapping identifies which levels fall in THz, GHz, or sub-GHz phonon regimes.

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1. Phonon Frequency Mapping

$$\omega_n = \frac{E_n}{\hbar} = \frac{E_0}{\hbar} \cdot (2\pi)^{n/3} \cdot S_{26}$$

2. Quality Factor

$$Q_n = \frac{\omega_n \cdot \sqrt{E_n}}{k_B T}$$

3. Regime Classification

Level Range	ω_n	Regime
n = 1-5	Sub-THz	Acoustic phonon
n = 6-15	THz	Optical phonon
n = 16-26	Multi-THz	High-frequency phonon

4. 26-Level Table

Each level n from 1 to 26 produces a unique (E_n, ω_n, Q_n) triplet, fully determined by the UQFF spectral ladder constants.

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5. Source Data

- **File:** et_{phonon_resonance}.py §7
- **Session:** 214
- **CP4 Class:** SpectralLadderPhononMappingCalc (#540)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_952 — 26-State HRes Spectral Ladder
3. PAPER_955 — BCS Phonon Resonance
4. PAPER_953 — Ramanujan-Accelerated S_{26}

Cross-Links

Related Paper	Relationship
PAPER_952	Energy levels E_n mapped here
PAPER_955	BCS Q-factor at each level
PAPER_959	Ramanujan summation driving S_{26}
PAPER_964	Magnetar phonon shells use this mapping

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi G M m_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ J/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> *Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\mathrm{SSq}]| < 1$ (satisfied by $[\mathrm{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\mathrm{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\mathrm{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day^{-1}	Damping rate
$[SSq]$	—	0.57	String coupling
E_0	$\hbar\omega_{\mathrm{SCm}}$	8.27×10^{-22} J	Ladder base

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
26 phonon frequencies	$\omega_n = E_n/\hbar$	Derived
Q-factor classification	narrow / optimal / broad	Validated

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Phonon Mapping (Spectral Ladder → Frequency)

§A.2 Lagrangian Density

$$\mathcal{L}_{\mathrm{map}} = \sum_{n=1}^{26} \left[\frac{1}{2}\hbar\omega_n \hat{a}_n^\dagger \hat{a}_n\right]$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\omega_n = \frac{E_0}{\hbar}(2\pi)^{n/3} S_{26}, \quad Q_n = \frac{\omega_n}{2\Gamma}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\to$ spectral ladder $\to E_n \to \omega_n$ phonon frequency $\to Q_n$ quality factor $\to$ regime classification

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Each ω_n traces a VDS density shell at radius $\propto n$.

§B.2 DVP

Prime-indexed levels ($n = 2, 3, 5, 7, 11, 13$) mark magnetic shell closures.

§B.3 BSH

Q_n saturation profile: $\mathrm{BSH}(n) = Q_n/Q_{\max}$.

§B.4 Production-Scale Consistency

Metric	Value	Status
26 levels	All mapped	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy

16 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
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